$A = B \cdot C$

2.5 Matrix Factorizations

- A factorization of a matrix A is an equation that expresses A as a product of two or more matrices.

The LU Factorization

- The LU factorization is motivated by the fairly common industrial and business problem of solving a sequence of equations, all with the same coefficients matrix.

$$Ax = b_1,\ Ax = b_2,\ \ldots,\ Ax = b_p$$

- At first, assume that A is an $m \times n$ matrix that can be row reduced to echelon form, without row interchanges (later we will treat the general case). Then, A can be written in the form $A = LU$, where L is an $m \times m$ lower triangular matrix with 1s on the diagonal and U is an $m \times n$ echelon form of A.

Ex:

$$A = \underset{4\times5}{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_{L} \underset{4\times4}{\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{U} \underset{4\times5}{\quad}$$

■ : any nonzero value

* : any value

An LU factorization of A.

The matrix L is invertible and is called a unit lower triangular matrix.

$Ax=b \rightarrow L(Ux)=b \Rightarrow Ly=b$

– Before studying how to construct L and U, We should look at why they are so useful. When $A = LU$, the equation $A\boldsymbol{x} = \boldsymbol{b}$ can be written $L(U\boldsymbol{x}) = \boldsymbol{b}$. Writing y for $U\boldsymbol{x}$, we can find $\boldsymbol{x}$ by solving the pair of equations:

$$L\boldsymbol{y} = \boldsymbol{b} \quad (1)$$
$$U\boldsymbol{x} = \boldsymbol{y} \quad (2)$$

– First solve $L\boldsymbol{y} = \boldsymbol{b}$ for $\boldsymbol{y}$ and then solve $U\boldsymbol{x} = \boldsymbol{y}$ for $\boldsymbol{x}$. Each equation is easy to solve because L and U are triangular.

Ex:

With this example we will try to show that solving a sequence of equations by using the LU factorization will be faster than applying the row reduction algorithm for each equation.

– Suppose were given the following LU factorization of a specific A matrix.

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & 5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}}_{U} = LU$$

Use this LU factorization of A to solve $Ax=b$ where $b=\begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$

The solution of $Ly = b$:

$$[L \quad b] = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 + R_1 \to \\ R_3 + -2R_1 \to \\ R_4 + 3R_1 \to \end{matrix}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ 0 & 8 & 3 & 1 & -16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ 0 & 8 & 3 & 1 & -16 \end{bmatrix}$$

E_1 $\qquad$ $[L \; b]$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ 0 & 8 & 3 & 1 & -16 \end{bmatrix} \xrightarrow{\begin{matrix} R_3 + 5R_2 \to \\ R_4 + -8R_2 \to \end{matrix}} \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 3 & 1 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & -8 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & -5 & 1 & 0 & 25 \\ 0 & 8 & 3 & 1 & -16 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 3 & 1 & 16 \end{bmatrix}$$

E_2

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -3 & 1 \end{bmatrix}$$

E_3

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 3 & 1 & 16 \end{bmatrix} = R_4 - 3R_3 \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = [I \; y]$$

6 addition
6 multiplication

$\therefore y = b$ ✓ when $y = \begin{bmatrix} -9 \\ -4 \\ 5 \\ 1 \end{bmatrix}$. Now solve $Ux = y$.

$$[U \; y] = \begin{bmatrix} 1 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} = R_4 * -1 \rightarrow \begin{bmatrix} 1 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\begin{matrix} R_1 + -2R_4 \rightarrow \\ R_2 + -2R_4 \rightarrow \\ R_3 + -1R_4 \rightarrow \end{matrix} \begin{bmatrix} 1 & -7 & -2 & 0 & -7 \\ 0 & -2 & -1 & 0 & -2 \\ 0 & 0 & -1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} = R_1 * -1 \rightarrow \begin{bmatrix} 1 & -7 & -2 & 0 & -7 \\ 0 & -2 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\begin{matrix} R_1 + 2R_3 \rightarrow \\ R_2 + R_3 \rightarrow \end{matrix} \begin{bmatrix} 1 & -7 & 0 & 0 & -19 \\ 0 & -2 & 0 & 0 & -8 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} = R_2 / -2 \rightarrow \begin{bmatrix} 1 & -7 & 0 & 0 & -19 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$R_1 + 7R_2 \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 9 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} = R_1 / 3 \rightarrow \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{I} \underbrace{\begin{bmatrix} 1 \\ 4 \\ -6 \\ -1 \end{bmatrix}}_{x}$$

$x = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix}$

(28) $Ly = b$ ① , $Ax = y$ ②

vs

(62) $Ax = b$

4 divisions
6 mult.
6 addition

Finding x requires 28 operations or "flops" (floating point operations) excluding the cost of finding L and U.

In contrast, row reduction of $[A\ b]$ to $[I\ x]$ takes 62 operations.

An LU Factorization Algorithm

1. Reduce A to an echelon form U by a sequence of row placement operations, if possible.
2. Place entries in L such that the same sequence of operations reduces L to I.

Ex: Find an LU factorization of

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

The row operations that create zeros in the first column A will also create zeros in the first column of L. Highlight the elements in each matrix that are used to determine the sequence of row operations that transform A into U.

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} \sim \begin{matrix} R_2 + 2R_1 \to \\ R_3 - R_1 \to \\ R_4 + 3R_1 \to \end{matrix} \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix}$$

$$= \begin{matrix} R_3 + 3R_2 \to \\ R_4 - 4R_2 \to \end{matrix} \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U$$

$R_1 - 2R_3$

The highlighted entries (entries in red rectangles) above determine the row reduction of A to U. At each pivot column, divide the highlighted entries by the pivot and place the result into L.

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} \quad \begin{bmatrix} 3 \\ -1 \\ 12 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad [5]$$

$/2 \quad /3 \quad /2 \quad /5$

$$\begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 1 & -1 & 1 & \\ -1 & 4 & 2 & 1 \end{bmatrix} = L$$

$A \overset{?}{=} LU$

An easy calculation verifies that $L \cdot U = A$. ✓

An explanation why $L \cdot U = A$

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

$R_2 + 2R_1 \to$
$R_3 - R_1$
$R_4 + 3R_1$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix} \cdot A =$$

E_1

$$= \underbrace{\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix}}_{E_1 \cdot A} \quad \begin{matrix} R_3 + 3R_2 \\ R_4 - 4R_2 \end{matrix} \equiv \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & -4 & 0 & 1 \end{bmatrix}}_{E_2} \cdot E_1 A$$

$$= \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \quad R_4 - 2R_3 \equiv \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix}}_{E_3} \cdot E_2 \cdot E_1 \cdot A$$

$$\underbrace{\qquad\qquad}_{E_2 \cdot E_1 \cdot A}$$

$$= \underbrace{\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}}_{E_3 \cdot E_2 \cdot E_1 A} = U.$$

$$E_3 \cdot E_2 \cdot E_1 \cdot A = U \Rightarrow A = (E_3 \cdot E_2 \cdot E_1)^{-1} \cdot U$$

$$A = L \cdot U$$

Then, $L = (E_3 \cdot E_2 \cdot E_1)^{-1}$

If you multiply $E_3 \cdot E_2 \cdot E_1$ and take its inverse you'll see that you obtain L, indeed.